

AlphaStock: Executive Presentation Report

1. Project Overview

AlphaStock is a full-stack, AI-driven quantitative analytics platform designed for the NIFTY 50. It transforms over a decade of raw market data into actionable trading intelligence, providing users with a professional-grade terminal for portfolio management and risk analysis.

2. Technology Stack (How We Built It)

We utilized a decoupled architecture to ensure maximum performance and scalability:

- Frontend (UI/UX):** Built with **React 19** and **Vite 8**, featuring a premium dark-mode interface styled with **TailwindCSS 3.4**. It includes offline-first resilience so the terminal functions even if the connection drops.
- Backend (API):** Engineered in **Python 3.12** using **FastAPI** for high-speed, asynchronous data delivery.
- Database:** **PostgreSQL** handles user accounts, watchlists, and secure preferences, integrated via **SQLAlchemy**.
- Infrastructure:** The application is fully containerized with **Docker** and deployed continuously via **Vercel** (Frontend) and **Render** (Backend).

3. Data Pipeline & Market Context

- Data Sourcing:** Historical pricing and volume ingested via `yfinance`; real-time market sentiment via `newsapi`.
- Feature Engineering:** We compute over 130 technical indicators (e.g., RSI, MACD, Volatility bands) using `pandas-ta`.
- Regime Detection:** We use **Hidden Markov Models (HMM)** to automatically classify the market into states (Bull, Bear, Sideways, Crisis), allowing models to adapt dynamically to market conditions.

4. Artificial Intelligence & Predictive Models

We discarded simple moving averages in favor of an institutional-grade machine learning pipeline:

- Tree-Based Ensembles (XGBoost & LightGBM):** Used for highly accurate, non-linear classification of directional movement (UP/DOWN) across 1-day, 5-day, and 20-day horizons.
- Temporal Fusion Transformers (TFTs):** A state-of-the-art deep learning model tailored for time-series forecasting. TFTs analyze past prices and static covariates (like sector) to output predictions with a strict *confidence interval*, essential for institutional risk management.
- Model Explainability (SHAP):** We use SHAP values to "open the black box." Every prediction includes an explanation (e.g., "Expected to rise due to oversold RSI and positive momentum"), ensuring complete transparency for analysts.

5. Portfolio Optimization & Backtesting

- Optimization:** Markowitz Mean-Variance optimization dynamically allocates asset weights based on the user's risk tolerance to maximize the Sharpe ratio.
- Backtesting:** A robust walk-forward simulation engine prevents data leakage, reporting key metrics (Max Drawdown, Calmar Ratio, Excess Return) against a Buy-and-Hold benchmark.

6. Future Roadmap (Advanced Risk Modeling)

To elevate AlphaStock to the next level of quantitative risk management, we are integrating:

- Merton Jump Diffusion (MJD) Models:** To account for sudden market shocks and "jumps" caused by unpredictable macroeconomic news.
- Monte Carlo Simulations:** To generate thousands of future price paths using the MJD framework, calculating accurate Value at Risk (VaR) and Expected Shortfall for extreme scenario planning.